Advisor Meeting Notes

Sudipta Poudel · Meeting with Dr. Pinliang Dong · October 30, 2026

1. Opening (30 seconds)

"Thank you for your suggestions last time, Dr. Dong. I followed them: I moved the study to Dallas, focused on urban green infrastructure, used NAIP and LiDAR instead of hyperspectral, and made GeoAI transferability the main question. I'd like to show you my progress and ask for your advice on a few decisions."

2. The Topic (1 minute)

Working title: *Geographic Transferability of Supervised Deep Learning and Foundation Models for Urban Tree Canopy Mapping Using NAIP and LiDAR in Dallas, Texas*

Main question: If a model maps tree canopy well in one type of urban area, does it still work in a different type, and can geography explain and predict where it fails?

Research questions

1. How accurate are U-Net, SAM 2 and SAM 3 for canopy mapping from NAIP, and how much does LiDAR help?
2. How much accuracy is lost when a model is applied to a different urban zone?
3. Which geographic characteristics explain that loss, and can a dissimilarity index predict where a model is reliable?

3. Why It Matters (1 minute)

"Dallas has about 29% canopy, but it is very uneven. Much of it is in the Great Trinity Forest, and some commercial areas have less than 5%. The city adopted an Urban Forest Master Plan in 2021, so reliable canopy monitoring is needed. Recent studies, such as Ventura et al. (2024) and Yoo et al. (2026), mapped NAIP canopy and tested transfer between cities, but they did not explain the loss by urban form, and SAM 3 has not been compared yet."

4. Design (1 minute)

"I use Local Climate Zones to define four contrasting urban types, one 1 km × 1 km square each. Models are trained or tuned in three zones and tested in the fourth (leave-one-zone-out), with 20 reference plots of 100 m × 100 m per zone."

5. Work Done So Far (2 minutes)

Literature

- A list of about 100 verified papers, and a full proposal draft.

Data (Phase 1 complete)

- Downloaded the City of Dallas boundary (Dallas Open Data, updated March 2026) and projected it to NAD83 UTM Zone 14N.
- Downloaded the Global LCZ map v3 (Demuzere et al.) and clipped it to Dallas. It matches the known structure: downtown, Uptown, the Trinity floodway and the industrial corridor.
- Chose four study areas objectively with a Python script (1 km grid, LCZ percentage per cell, then a visual check on imagery).

Zone	LCZ	Cell ID	Purity	Place
Compact	1–2	437	88%	Downtown Dallas
Open low-rise	6	617	100%	North Dallas residential
Large low-rise	8	637	99.9%	Northwest industrial
Trees	A–B	352	98%	Great Trinity Forest

SAM 3 setup

- ArcGIS Pro 3.7 (Advanced) with Image Analyst; Deep Learning Libraries installed.
- Laptop GPU: NVIDIA RTX 5060 with 8 GB.
- Found “Segment Anything Model 3 (SAM3)” on ArcGIS Living Atlas.
- Next: test it on a small NAIP image from North Dallas with the prompt “tree”.

6. Questions to Ask

1. **Scope:** Should I map canopy cover only, or also individual trees?
2. **Study areas:** Are four zones enough, or should I add one, for example Oak Cliff (lower canopy), for an equity angle?
3. **Models:** Is comparing U-Net, SAM 2 and SAM 3 good? Should I also include Esri’s Tree Segmentation model from Living Atlas as a baseline?
4. **LiDAR:** Dallas has 2019 LiDAR and possibly 2025 LiDAR. Is a few years’ gap with NAIP 2022 acceptable if I exclude plots that changed?
5. **Reference data:** Is digitizing 80 plots of 1 hectare reasonable, or should I reduce it?
6. **GNN:** I am interested in using a graph neural network to refine SAM segments with LiDAR and NDVI. Should it be part of the thesis, or future work?

7. **Computing:** Does the lab have a GPU workstation I could use if my laptop is not enough?
8. **License:** My ArcGIS license expires December 9, 2026. Will UNT renew it automatically?

7. Closing (30 seconds)

“Based on your feedback, my next steps are to test SAM 3 on a small NAIP image, download NAIP and LiDAR for the four study areas, and update the proposal. Would it be okay to send you the updated proposal after that?”

Notes from the Meeting